Dual-Branch Melanoma Classification with Metadata Fusion, Focal Loss, and Mahalanobis OOD Detection

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***Abstract*—This work presents a dual-branch deep learning system fusing dermoscopic images with patient metadata (age, sex, anatomical site) for binary melanoma classification on the SIIM-ISIC 2020 dataset (13.6% malignant). Using an EfficientNet-B0 backbone with sigmoid focal loss (α=0.864), a hyperparameter search across five experiments (Baseline, Experiments 1–4) isolates the effect of learning-rate scheduling, optimiser choice, and focal γ. Cosine annealing emerges as the most impactful change (F1: $0.8932 \rightarrow 0.8984$, false negatives reduced 12.5%). An image-only ablation (Experiment 4) confirms metadata fusion adds +1.9 percentage points (pp) recall; SHAP analysis identifies anatomical site and age as dominant predictors. With six-transform test-time augmentation (TTA) the system achieves F1 = 0.9134, recall = 91.48%, specificity = 98.62% (two-transform eval TTA on Baseline; Gradio demo uses six transforms). EigenCAM validates ABCDE-aligned attention and a Mahalanobis OOD detector flags non-dermoscopic inputs.**

***Index Terms*—melanoma, focal loss, metadata fusion, SHAP, test-time augmentation**

I. Introduction

Melanoma causes over 75% of skin cancer deaths despite representing fewer than 5% of cases [1]. Five-year survival exceeds 98% at stage I but falls below 25% for metastatic disease [2]. Dermoscopy improves sensitivity by 10–27 pp [3], yet inter-observer variability persists. Esteva et al. [4] showed a fine-tuned CNN matched 21 dermatologists, and the ISIC 2020 challenge [6] released structured metadata alongside dermoscopic images, motivating late-fusion approaches that incorporate patient context. A further challenge is severe class imbalance: at ∼13.6% positive rate, a trivial all-benign predictor achieves 86.4% accuracy [11].

We select **EfficientNet-B0** [8] as the backbone because (i) the ISIC 2020 winning solution used an EfficientNet ensemble [7], (ii) compound scaling offers a strong accuracy-per-FLOP trade-off at only 5.3 M parameters, and (iii) its smooth convergence makes it well-suited for controlled hyperparameter (HP) studies (Section IV-C).

Contributions:

1) Dual-branch fusion (CNN + metadata MLP) with focal loss α=0.864 derived from class proportions.
2) Five-experiment HP search (Baseline, Exp. 1–4) isolating scheduler, optimiser, focal γ, and metadata fusion, with per-experiment analysis of false-negative / false-positive trade-offs.
3) Measured ablation (+1.9 pp recall from metadata), per-feature SHAP analysis, error analysis, EigenCAM saliency, and Mahalanobis OOD.

Out of scope: Vision Transformers [17], [18] (require more data to compensate for lack of CNN inductive biases [17]), GAN augmentation, ensembles, multi-backbone comparison, multi-class diagnosis, and the ISIC private test set.

II. Literature Review

A. CNN Backbones for Dermoscopy

The ISIC 2020 challenge [6] released 33,126 dermoscopic images with patient metadata; our training set concatenates ISIC 2020 with additional ISIC 2019 images, yielding 37,648 samples. The winning solution used an EfficientNet ensemble [7] (AUC = 0.9490). **EfficientNet-B0** [8] applies compound scaling to balance accuracy and efficiency with only 5.3 M parameters. DenseNet-121 [9] and ResNet-50 [10] are popular alternatives but require more parameters for equivalent accuracy. ViTs [17], [18] show promise but need substantially more data due to lack of translation equivariance [17]; CNNs with ImageNet pretraining remain pragmatic at this scale.

B. Class Imbalance and Focal Loss

With 6.4 benign examples per malignant, standard cross-entropy steers gradients toward the majority class so the network can ignore rare positives and still minimise loss. Focal loss [11] multiplies the per-example loss by $(1 - p_t)^\gamma$: confident, correct predictions contribute nearly zero gradient; uncertain or wrong predictions receive amplified signal, forcing attention onto hard malignant cases. Setting α=1−0.136=0.864 further upweights malignant examples. Effect of varying γ is in Section IV-B.

C. Metadata Fusion, TTA, and Limitations

Pacheco et al. [12] curate a multimodal skin lesion dataset pairing clinical images with patient metadata, motivating fusion approaches. Groh et al. [13] caution about skin-tone bias. TTA [14] averages predictions over augmented views to reduce variance. Fujisawa et al. [15] identify amelanotic and small lesions as persistent DL failure modes. Davis and Goadrich [16] show AUC can overstate clinical utility on imbalanced data, motivating per-category error analysis (Section IV-E).

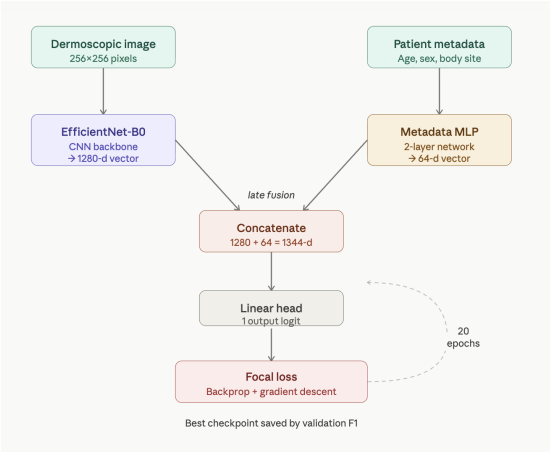

Fig. 1. Dual-branch pipeline: EfficientNet-B0 image branch and metadata MLP are fused via late concatenation, trained end-to-end with focal loss.

III. Methodology

A. Dataset

The SIIM-ISIC 2020 dataset contains ∼37 k dermoscopic images (256×256, **13.6% malignant**) with metadata: `age_approx` (continuous), `sex` (binary), and `anatom_site` (6 categories). An 80/20 stratified split (seed 42) preserves the class ratio. All experiments share this identical split.

B. Preprocessing and Augmentation

Images (train): Albumentations (CLAHE, blur, 90° snaps) then torchvision v2 (flip, affine ±15°, colour jitter, random erasing, ImageNet normalisation). **Images (val):** resize + normalise only. **Metadata:** median imputation + z-score for age; mode imputation + one-hot for categoricals → **14-d** vector.

C. Architecture

A dual-branch late-fusion model (Fig. 1): (1) an ImageNet-pretrained **EfficientNet-B0** backbone (→ 1280-d embedding) with the classification head removed; (2) a two-layer metadata MLP (14→128→64, BatchNorm + ReLU + Dropout 0.3). Late fusion is chosen because images and clinical scalars benefit from separate processing before being combined. The 1280-d and 64-d embeddings are concatenated (→ 1344-d) and projected to a single logit. Exp. 4 disables the metadata branch to isolate its contribution.

D. Loss and Training

Sigmoid focal loss [11]:

$$FL(p_t) = -\alpha_t \, (1 - p_t)^{\gamma} \log(p_t) \tag{1}$$

with α=0.864, γ=2.0. We define five experiments (Table I), each introducing *one primary change* so every factor can be isolated: **Baseline** (Adam, fixed LR, γ=2.0): minimal-tricks reference. **Exp. 1** adds cosine annealing to settle into sharper minima (higher F1, no new FP). **Exp. 2** Adam→AdamW + weight decay 10^{-5} to curb minority-class overfitting. **Exp. 3** lowers γ to 1.5, reducing suppression of easy negatives to recover specificity lost when Exp. 2 widened the decision boundary. **Exp. 4** disables metadata MLP (image-only ablation). The checkpoint with the highest validation F1 is retained.

TABLE I
HP configurations and best-checkpoint results (EfficientNet-B0). All runs: LR=10^{-4}, batch 32, 20 epochs; WD=10^{-5} for Exp. 2–3, else 0. ΔFN: change in false negatives vs. prior row; ‡vs. Baseline.

Config	Optim	Sched	γ	Meta	F1	Recall	FN (ΔFN)	FP
Baseline	Adam	None	2.0	✓	0.8932	92.16%	80 (–)	145
Exp. 1 (cosine)	Adam	Cosine	2.0	✓	**0.8984**	93.14%	70 (−10)	145
Exp. 2 (AdamW)	AdamW	Cosine	2.0	✓	0.8855	**93.54%**	66 (−4)	181
Exp. 3 (γ1.5)	AdamW	Cosine	1.5	✓	0.8968	92.75%	74 (+8)	144
Exp. 4† (no meta)	Adam	None	2.0	—	0.9000	90.30%	99 (+19)‡	106

†Image-only ablation (no metadata MLP).

E. Test-Time Augmentation

During evaluation, two geometric transforms (original and horizontal flip) are averaged to reduce orientation-specific variance. The Gradio demo extends this to six transforms (original, h-flip, v-flip, 90°/180°/270° rotation), exploiting the rotational symmetry of dermoscopic lesions. Evaluation-pipeline TTA impact is reported in Section IV-A.

F. Evaluation

Five metrics are reported on the validation split: **Recall** $= TP/(TP{+}FN)$: of all real cancers, how many did we catch. **Specificity** $= TN/(TN{+}FP)$: of all benign cases, how many were correctly cleared. **PPV** $= TP/(TP{+}FP)$: of everything flagged as cancer, how many actually were. **F1**: harmonic mean of PPV and recall—penalises models that sacrifice either. **Accuracy** is reported but not optimised: predicting benign always yields 86.4% accuracy yet catches zero cancers, making it clinically meaningless on this dataset. Recall is the primary metric: a missed melanoma carries substantially higher clinical cost than a false alarm.

IV. Experiments

All experiments use PyTorch 2.x with the EfficientNet-B0 backbone fine-tuned for up to 20 epochs. Identical data splits, augmentation pipelines, and evaluation code are shared across all configurations.

A. Incremental Ablation

Exp. 4 (image-only, same HP as Baseline) isolates the metadata branch. Its accuracy (97.28%) and raw F1 (0.9000) look strong, but the confusion matrix reveals the cost: 99 missed melanomas versus Baseline's 80—19 more patients undetected. At 13.6% prevalence, 19 extra misses barely register across 7,530 examples, illustrating why accuracy overstates clinical utility on imbalanced data [16]. Metadata adds +1.9 pp recall (FN: 99→80) at the cost of 39 extra FP, because the MLP responds to anatomical-site cues absent from image features alone. Fig. 2 extends this comparison to the best

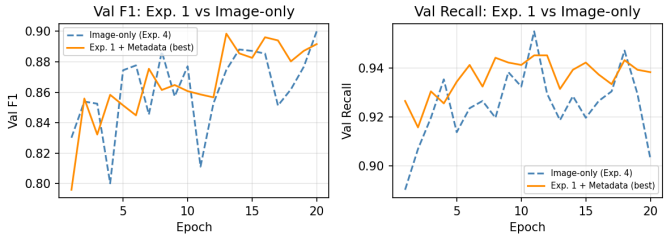

Fig. 2. Validation F1 and recall over 20 epochs: image-only (Exp. 4) vs. Exp. 1 (best model, with metadata MLP + cosine annealing). Metadata fusion provides a consistent recall advantage throughout training.

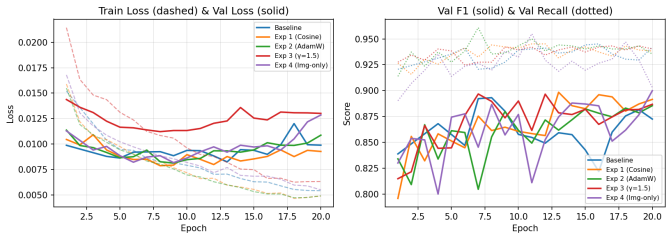

Fig. 3. Training and validation loss (left) and validation F1/recall (right) for all five experiments. Cosine annealing (Exp. 1–3) drives training loss lower; Exp. 4 (image-only) tracks Baseline but with lower F1, confirming the metadata contribution.

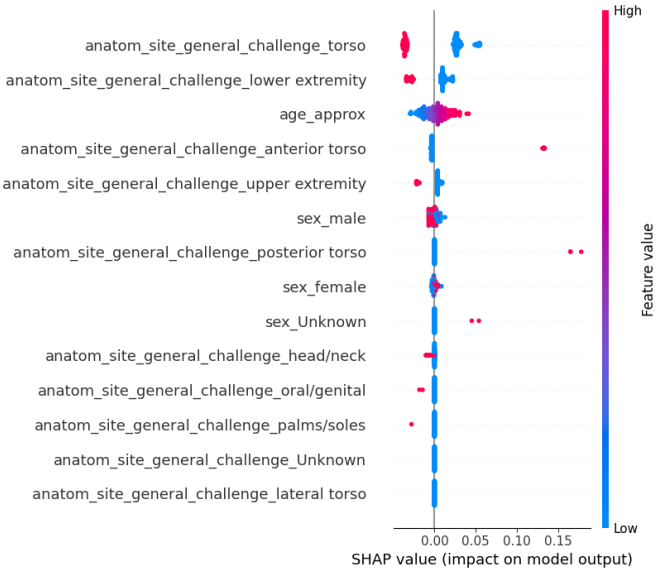

Fig. 4. SHAP beeswarm plot for Exp. 1 (best model). Feature importance is ranked by mean |SHAP value|; colour indicates feature magnitude (red = high, blue = low). `anatom_site_torso` and `age_approx` are the dominant predictors.

model (Exp. 1), confirming that Exp. 1 maintains a consistent recall lead over the image-only ablation across all 20 epochs.

Two-transform TTA (original + h-flip) applied to the Baseline model trades 7 extra FN for 55 fewer FP (recall 92.16%→91.48%, F1 = 0.9134, specificity 98.62%), tightening the decision boundary at a small sensitivity cost.

B. Hyperparameter Search

The five experiments form a deliberate chain (Table I). **Exp. 1** (cosine annealing): FN drops 80→70 with *zero* extra FP (ΔF1 = +0.0052) — the strongest single improvement. **Exp. 2** (AdamW): peaks recall at 93.54% but surges FP 145→181 (ΔF1 = −0.0129), broadening the decision boundary. **Exp. 3** (γ=1.5): restores specificity — FP falls to 144 (ΔF1 = +0.0113) while FN rises modestly to 74. Metadata fusion (Baseline vs. Exp. 4) adds +1.9 pp recall (19 fewer missed cancers) at the cost of 39 extra FP; the ΔF1 of −0.0068 reflects increased FP, not worse cancer detection. Two-transform TTA on the Baseline adds ΔF1 = +0.0202, trading 7 extra FN for 55 fewer FP.

C. Training Dynamics

Fig. 3 overlays training dynamics. Baseline converges smoothly (loss 0.0152→0.0054, gap <0.32%). Cosine annealing (Exp. 1–3) drives loss lower but widens the gap (0.36–0.59%), reflecting focal loss sensitivity to class boundaries on an imbalanced dataset; best-checkpoint selection by validation F1 ensures reported results are not sensitive to epoch-to-epoch variance. Exp. 3 (γ=1.5) shows the largest gap; Exp. 1 achieves the best trade-off.

All five experiments achieve AUC = 0.99, confirming HP variations affect the precision–recall operating point rather than ranking ability [16].

D. SHAP Feature Importance

SHAP KernelExplainer (Fig. 4) reveals the metadata feature importance for Exp. 1 (best model). `anatom_site_torso` is the dominant predictor; `age_approx` shows a monotonic relationship (older → higher risk); `anterior_torso` is rare but high-impact (SHAP up to 0.18). Sex contributes modestly. Feature rankings are consistent across all configurations; Exp. 3 (γ=1.5) is the only exception, compressing magnitudes and distributing importance more evenly.

E. Error Analysis

FN (∼7–8.5% of malignant): amelanotic melanomas, small lesions (<6 mm), and early superficial spreading variants—subtypes lacking the pigmented features CNNs rely on [15]. **FP (∼1.4–2.8% of benign):** dysplastic naevi, atypical Spitz naevi, and irregular seborrhoeic keratoses [5]. Exp. 1 achieves a 6.86% FN rate, acceptable for a screening tool.

F. OOD Robustness

A Mahalanobis distance-based OOD detector [19] computes μ and Σ^{-1} of the backbone's 1280-d validation features; inputs beyond $\mu+3\sigma$ are flagged. Thresholds range from 2637 (Exp. 3) to 3248 (Baseline); cosine-annealed models produce tighter feature distributions. Without this gate, non-dermoscopic inputs activate similar low-level CNN filters and align with malignant-class weights, yielding misleadingly high confidence—Fig. 5 demonstrates a koala scoring distance 5124 ($\gg$ threshold 2652) yet receiving 85.8% malignant from the

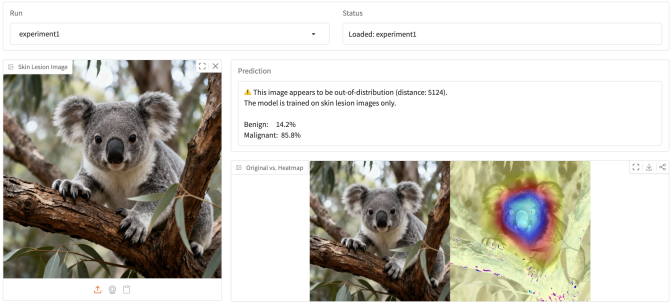

Fig. 5. OOD detection in the Gradio demo (Exp. 1). A koala photograph yields Mahalanobis distance 5124, far exceeding the threshold of 2652. The system issues an OOD warning; without it the unconstrained sigmoid head assigns 85.8% malignant confidence.

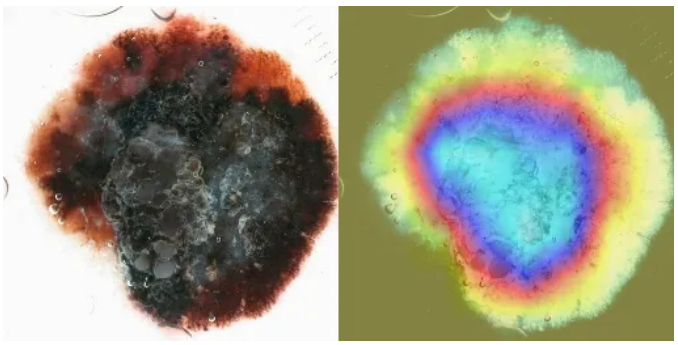

Fig. 6. EigenCAM overlay for a malignant lesion (Exp. 1). Left: original dermoscopic image. Right: activation heatmap—attention concentrates on asymmetric border and colour variegation, consistent with clinical ABCDE criteria [4].

raw sigmoid. The detector intercepts at the embedding level before the logit is trusted.

G. EigenCAM Interpretability

EigenCAM (Fig. 6) saliency maps confirm attention localises to asymmetric borders, colour variegation, and atypical pigment networks, aligning with clinical ABCDE criteria [4].

V. Conclusion and Future Work

A dual-branch EfficientNet-B0 melanoma classifier fusing dermoscopic images with patient metadata achieves F1 = 0.8984, recall = 93.14% (Exp. 1, best epoch 13), reducing missed melanomas by 12.5% over Baseline with no additional FP. Two-transform TTA (original + h-flip) on the Baseline achieves F1 = 0.9134, recall = 91.48%, specificity = 98.62% (trades 0.7 pp recall for 55 fewer FP; the Gradio demo extends this to six transforms). Ablation (Exp. 4) confirms metadata adds +1.9 pp recall; SHAP identifies `anatom_site_torso` and `age_approx` as dominant predictors across all HP experiments. EigenCAM validates ABCDE-aligned attention. A Mahalanobis OOD detector with empirically computed thresholds provides runtime safeguards against non-dermoscopic inputs.

Deployment. The full codebase is open-sourced at https://github.com/burrows99/melanoma-classification. Trained weights for all five experiments are hosted at https://huggingface.co/burrows99/melanoma-models, and a public Gradio demo is deployed at https://huggingface.co/spaces/burrows99/melanoma-classificiation, enabling browser-based inference with EigenCAM overlays and OOD detection.

Future work: (1) Larger EfficientNet variants (B3/B5) [7]; (2) multi-backbone comparison (DenseNet-121 [9], ResNet-50 [10]); (3) equity assessment via Fitzpatrick stratification [13]; (4) external validation on the ISIC private test set; (5) ViT/Swin backbones [17], [18] with larger pretraining budgets.

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